

BayCare Digital Health Maturity Assessment

Industry: Health Care Delivery

Selected Organization: BayCare Health System

Use Note: Independent educational assessment; not an official BayCare evaluation

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Individual GitHub Pages URLs

https://chandrahas-reddy-bheemireddy.github.io/ISM6930_005/week5/

<https://nikhitharediee12.github.io/maturity-assessment-tool/>

<https://swapnatumma.github.io/Team-maturity-assessment-tool/>

<https://tanishputtagunta.github.io/BayCare-Maturity-Assesment/>

Question Design Rationale

This assessment measures digital health maturity for BayCare Health System, a large, integrated care organization whose services span hospitals, ambulatory locations, laboratories, imaging, home care, virtual care, and patient-facing digital channels. The five dimensions are Patient Data & Interoperability, Digital Patient Experience, Clinical & Administrative Automation, AI, Analytics & Virtual Care, and Leadership, Governance & Workforce Adoption. Together they test whether technology is supported by dependable data, redesigned workflows, measurable outcomes, responsible governance, and sustained human adoption. The dimensions are intentionally distinct: a health system can offer an excellent app while its records remain difficult to reconcile, or deploy advanced technology while clinicians maintain old workarounds.

The four Patient Data & Interoperability questions examine cross-setting data exchange, consistent identities and terminology, governed use of unstructured data, and data-quality, privacy, and access controls. These questions distinguish manual reconciliation from a secure, standards-based, reusable data foundation. Evidence includes architecture diagrams, interface inventories, data dictionaries, stewardship assignments, quality dashboards, access reviews, and exception logs. The indicators are specific to health care because they address patient, provider, encounter, clinical-note, imaging, device, consent, and terminology data.

The Digital Patient Experience questions measure end-to-end digital completion, continuity between digital and human channels, inclusive access, and journey-level outcomes. Having a website or mobile application alone does not demonstrate maturity. Higher stages require patients' context to follow them across the BayCare app, portal, phone, virtual care, and in-person settings; accessibility and language needs to be tested; and improvements to be evaluated against access, adherence, resolution, and clinical follow-through. Percentage bands make the first question repeatable, while the remaining questions identify observable practices and records.

Clinical & Administrative Automation focuses on referrals, authorizations, care transitions, routine approvals, clinical burden, safety, and portfolio governance. The rubric does not equate installing workflow software with transformation. Stage 3 requires end-to-end ownership, documented rules, standard exceptions, human escalation, and baseline measures. Stages 4 and 5 add integrated automation, real-time outcome monitoring, human-factors evidence, and continuous redesign. AI, Analytics & Virtual Care follows the same logic. A pilot receives a lower score than a production service with clinical and technical owners, representative evaluation, monitoring, escalation protocols, and a documented decision to scale, revise, or stop.

Leadership, Governance & Workforce Adoption tests visible sponsorship, decision rights, proficiency, adoption, and psychological safety. These factors are critical because technical correctness does not guarantee safe use in clinical work. A defensible Stage 3 score requires evidence such as fixed leadership reviews, governance charters, role-based training, usage measures, and protected reporting channels - not statements such as "we have good people." Across all dimensions, the five options progress monotonically from ad hoc work to continuous optimization. Each level adds observable repeatability, ownership, measurement, control, or learning, enabling two assessors to review the same evidence and reach a similar result.

Interpretation Guide

Five archetypal score patterns translate results into a specific strategic next action.

Profile 1 - Foundation First

Pattern: All five dimensions score between 1 and 2.

BayCare may be funding isolated tools without the shared data, workflow ownership, safety controls, or repeatable change system needed to scale them.

Next action: Select one executive- and clinical-owner patient journey, map it end to end, and establish the data, workflow, safety, and adoption foundation needed to improve it.

Profile 2 - Connected Care Leader

Pattern: All five dimensions score between 4 and 5.

Digital capability is measured, governed, and embedded across care delivery. The challenge is sustaining innovation without allowing mature governance to slow responsible experimentation.

Next action: Create a controlled frontier portfolio for higher-risk AI, virtual-care, and patient-experience experiments with explicit safety, equity, value, and sunset criteria.

Profile 3 - Digital Front Door, Fragile Data

Pattern: Patient Data & Interoperability is significantly lower than the other dimensions.

Patient-facing services and automation are advancing faster than the data foundation beneath them, limiting safe personalization, analytics, and automation.

Next action: Fund an enterprise patient-data and interoperability product with named stewards, terminology standards, quality service levels, and a plan to retire duplicate interfaces.

Profile 4 - Access Without Intelligence

Pattern: Digital Patient Experience is high while AI, Analytics & Virtual Care is low.

Digital access is strong, but interaction and clinical data are not yet producing better decisions, proactive care, or integrated virtual services.

Next action: Choose one high-volume digital journey and deploy a governed analytics or AI use case tied to access, adherence, resolution, safety, or clinical outcomes.

Profile 5 - Capability Without Adoption

Pattern: Leadership, Governance & Workforce Adoption is significantly lower than the technical dimensions.

Platforms may launch while clinicians and staff retain old workarounds or inconsistent local practices. This is an execution and safety gap, not simply a technology gap.

Next action: Make executive, clinical, and operational leaders accountable for adoption outcomes, provide role-based support, and embed change leaders in the highest-friction workflows.

Stage Definitions

Each stage describes an observable health-system state. The progression is monotonic: every stage adds repeatability, ownership, measurement, control, or continuous improvement.

Patient Data & Interoperability

Stage 1 - Ad Hoc: Patient and operational data remain fragmented across systems and locations. Staff reconcile identities, definitions, access, and quality problems manually when needs arise.

Stage 2 - Developing: Major systems exchange selected data, and teams document some definitions. Point-to-point interfaces, duplicate records, and inconsistent remediation remain common.

Stage 3 - Defined: Priority settings share a longitudinal record through documented interfaces. Enterprise identifiers, terminology standards, stewards, access rules, and scheduled quality reporting exist for critical data.

Stage 4 - Managed: Standards-based connections provide timely data with lineage, automated validation, monitored access, service levels, and managed exceptions. Leaders use evidence to direct remediation.

Stage 5 - Optimizing: Interoperability operates as a reusable enterprise product. Preventive controls, automated policy enforcement, and continuous improvement support secure real-time use of structured and unstructured data.

Digital Patient Experience

Stage 1 - Ad Hoc: Most journeys depend on phone, paper, or in-person work. Channels operate separately, and patients repeat information.

Stage 2 - Developing: Several journeys are available digitally, but manual follow-up and disconnected context remain common. Accessibility is reviewed mainly during new launches.

Stage 3 - Defined: Priority journeys are digital end to end and use shared identity and case context. Accessibility, language, caregiver, and usability testing have named owners.

Stage 4 - Managed: BayCare measures completion, abandonment, accessibility, handoffs, response, resolution, and clinical follow-through across channels. Teams test improvements against outcomes.

Stage 5 - Optimizing: Patients move seamlessly across inclusive channels while teams continuously optimize personalized next actions and monitor safety, equity, experience, and clinical results.

Clinical & Administrative Automation

Stage 1 - Ad Hoc: Work moves through calls, faxes, email, paper, and spreadsheets without end-to-end ownership. Routine decisions and exceptions depend on individual effort.

Stage 2 - Developing: Departments automate local tasks, but cross-setting handoffs, audit trails, burden measures, and outcomes remain fragmented.

Stage 3 - Defined: Priority workflows have end-to-end owners, documented rules, standard exceptions, human escalation, safety review, audit trails, and baseline performance measures.

Stage 4 - Managed: Integrated workflows automate routine handoffs and decisions while dashboards monitor cycle time, burden, safety, exceptions, quality, and realized value.

Stage 5 - Optimizing: Human-factors evidence, process mining, predictive monitoring, and controlled experiments continuously improve workflows. Investment shifts toward demonstrated value and reusable components.

AI, Analytics & Virtual Care

Stage 1 - Ad Hoc: AI, analytics, remote monitoring, and virtual-care use is limited to isolated experiments with no shared evaluation, ownership, or portfolio process.

Stage 2 - Developing: Selected pilots operate, but workflow integration, clinical validation, escalation, outcome measures, and stop decisions are inconsistent.

Stage 3 - Defined: Production use cases have clinical, operational, and technical owners; documented evaluations; approved tools; risk tiers; expected outcomes; and standard escalation protocols.

Stage 4 - Managed: Multiple service lines use monitored capabilities with evidence on quality, safety, drift, bias, cost, access, outcomes, and human oversight. Stage gates determine whether work scales or stops.

Stage 5 - Optimizing: AI evaluation, analytics, virtual-care orchestration, and reusable platforms operate as enterprise products. Funding continuously shifts toward validated patient and organizational value.

Leadership, Governance & Workforce Adoption

Stage 1 - Ad Hoc: Digital change is delegated to IT, and facilities or functions make independent decisions. Training occurs at launch, while team members avoid raising concerns or testing new practices.

Stage 2 - Developing: Sponsors and committees exist, but mandates, participation, clinical ownership, and adoption responsibilities are inconsistent. Completion is measured more often than proficiency.

Stage 3 - Defined: Executive and clinical leaders review outcomes on a fixed cadence, decision rights and escalation paths are documented, and role-based support continues after launch. Protected reporting and low-risk experimentation exist.

Stage 4 - Managed: Leaders spend visible time with affected teams, managers are measured on adoption and safety outcomes, and governance performance is monitored. Low-adoption teams receive targeted support.

Stage 5 - Optimizing: Leadership behavior, incentives, learning, just-culture practices, psychological safety, and federated governance reinforce continuous change within monitored safety, privacy, and architecture guardrails.